

Which Wordification Matters? A Nineteen-Recipe Sweep of the Interpretable-Topic-Model Framework on Hyperspectral Imagery

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CAOS open-research programme · CAOS_LDA_HSI, topic models on hyperspectral imagery (paper P3)

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Not peer reviewed. Source code, data and computational records: github.com/fsantibanezleal/CAOS_LDA_HSI

Abstract. The companion paper [1] introduces a twelve-axis evaluation framework for topic models on hyperspectral imagery and instantiates it on a single canonical wordification recipe (V1, band-frequency tokenisation). This paper asks a different question: *how does the choice of wordification itself shape the conclusions of that framework.* We define a family of nineteen wordification recipes V1–V15, V17–V20 spanning seven axes of design freedom (token alphabet, spatial vs. spectral aggregation, local vs. global vocabulary, document-length regime, signal transform, label-aware weighting, and learnt representation) and sweep them through the framework on the six labelled-scene panel (Indian Pines [2], Salinas and Salinas-A [3], Pavia University [4], Kennedy Space Center [5] and Botswana [6]). The headline finding is that *there is no universal winner*: V1 wins F-2 coherence on 2/6 scenes; V3, a joint (band, bin) Cartesian vocabulary, wins F-7 topic-label normalised MI on 2/6 scenes; V12 (Gaussian-mixture tokens) wins F-2 on 2/6 and F-7 on 2/6; V20 (mutual-information-weighted bands, whose weights come from the labels F-7 is scored against) wins F-2 coherence (0.88) and F-7 NMI (0.44) together on Indian Pines; F-1 macro-F1 separates the recipes by less than 0.01 in six-scene mean (0.912 to 0.922; paired over scenes and folds, V12 leads every other recipe by 0.004 to 0.009), and V20’s F-1 is affected by label leakage through its MI weighting, as caveated in the F-1 protocol. V20 has the highest counterfactual robustness F-22 of the sweep at $Q = 8$ ($L_1 = 26.3$ vs 24.5 for V12), a token count that grows with its documents, about six times longer than V3’s. The full 19-recipe Q -sweep at $Q \in \{8, 16, 32\}$ identifies three recipes whose F-7 NMI and F-2 c_v both improve monotonically: V20, V2 (intensity-bin), and V8 (NFINDR endmember). V20 has the largest F-2 gain of the three (+0.060) and gains +0.043 on F-7 (V2: +0.045); it reaches the highest absolute F-7 value at $Q = 32$ (0.563); on F-2 c_v it ties V12 at $Q = 32$ (0.910 vs 0.909, margin 0.0016, a 3–3 per-scene split). At $Q = 8$ V20 trails V12 on the LDA F-7 mean by 0.014 NMI; the ranking *inverts* at $Q = 32$ where V20 leads the LDA F-7 mean by 0.030 and outscores V12 on 5/6 scenes (it wins F-7 *outright* on 4/6: Salinas-A, Pavia U, KSC and Botswana). Of the other fifteen recipes, seven decline monotonically on F-7 and four peak at $Q = 16$.

then regress. A four-backbone $\times$ nineteen-recipe extension of F-7 NMI identifies V8 (NFINDR endmember-fraction) as the most cross-backbone-consistent recipe by a clear margin (mean 0.431); V20 (0.397) and V2 (0.395) rank second and third, with V20 the only *label-aware* recipe in the consistently-strong family. The per-backbone F-7 winners are V12 (LDA, 0.534), V11 (HDP, 0.571), V8 (ProdLDA, 0.328) and V11 (ETM, 0.530). V13 (vector-quantised VAE) underperforms sharply; V18 (graph-Laplacian eigenvector tokens) places third on F-7 mean among the extension recipes (V13–V20). The per-axis winner depends both on the scene and on which property of the topic basis the axis measures. We report the nineteen-recipe result matrix on the six labelled scenes for F-1, F-2 and F-7, with the extension axes F-13, F-14, F-17, F-18 and F-22, identify four axis–recipe affinities, and recommend that V1 retain its canonical status as a *reproducibility default* rather than as a universal best. V12, V3 and V20 have the three highest six-scene means on both F-2 coherence and F-7 normalised MI, and on F-7 they lie within 0.014 NMI of each other, suggesting that the design-space ceiling for fixed-K LDA on the six labelled scenes is close to saturation; a foundation-model wordification (V16) is left to a separate study.

Index Terms. hyperspectral imagery, latent Dirichlet allocation, wordification, topic models, evaluation framework, interpretability, reproducibility, band-frequency tokens, absorption-feature tokens, product quantisation, Gaussian-mixture tokens, vector-quantised VAE, graph-Laplacian eigenvectors, UMAP coordinate tokens, mutual-information-weighted tokens, region-SAM tokens, comparison study.

I. INTRODUCTION

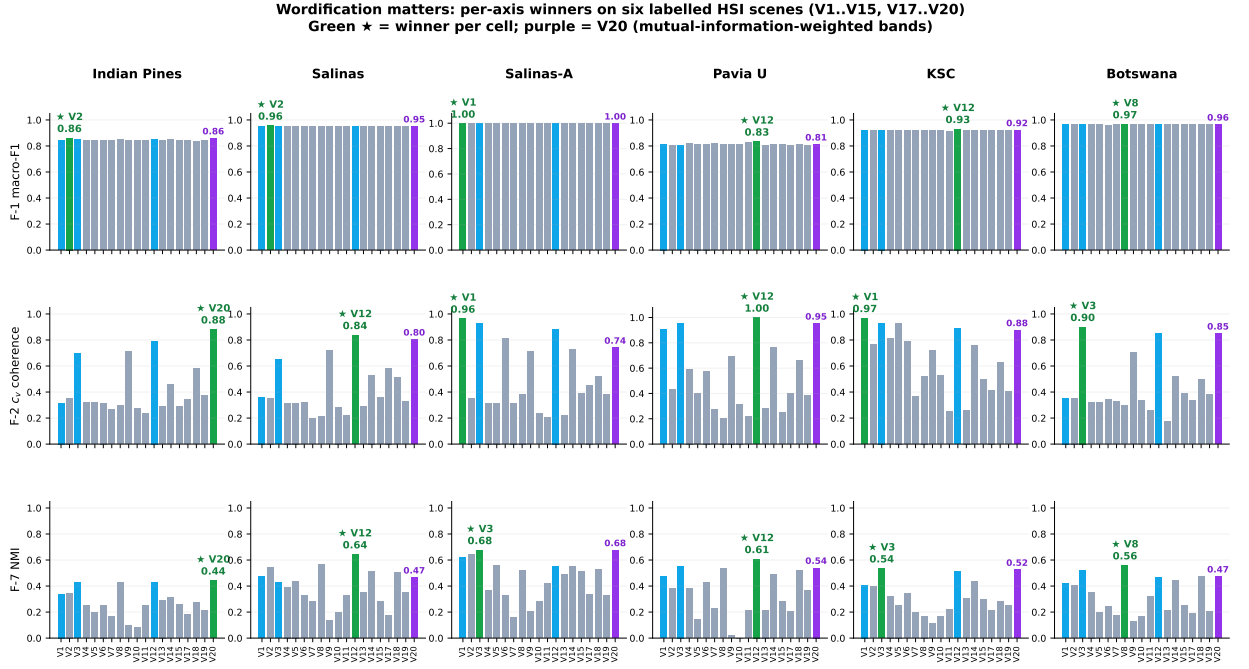
The companion paper [1] (denoted P1 hereafter) proposes a twelve-axis evaluation framework for LDA-style topic models on hyperspectral imagery, instantiates it on a single *canonical* wordification recipe (V1, band–frequency tokens), and shows that classification accuracy alone is silent about reproducibility, band robustness, and calibration. P1’s choice of V1 as canonical was a deliberate design decision: every axis pulls a property of the same basis ϕ , which in turn requires that ϕ be fixed across axes. The decision was defensible but it leaves an obvious second question unanswered: *does the choice of wordification itself change the conclusions?*

This paper answers that question. We define a family of nineteen wordification recipes V1–V15 and V17–V20 that span the axes of design freedom identified by the existing

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The twelve-axis evaluation framework, the dataset panel and the canonical V1 fit are those of the companion framework paper, referred to as P1 throughout. Code, derived artefacts, and the interactive web application: https://github.com/fsantibanezleal/CAOS_LDA_HSI.

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Indian Pines: V20 wins F-2 (0.88) and F-7 (0.44); on F-1 V2 leads (V2 0.861, V20 0.858). F-7 mean ranking: V12 0.534 / V3 0.524 / V20 0.520 (within 0.014 NMI of each other). Source: data/derived/v_sweep/{f1_per_fold,f2_coherence,f7_topic_to_label}/ in CAOS_LDA_HSI.

Fig. 1. Headline. Per-axis winners on six labelled HSI scenes for F-1 (5-fold macro-F1), F-2 (c_v coherence) and F-7 (topic-label NMI). Bars: green ★ = winner per cell; purple = V20 (MI-weighted bands). Indian Pines (left column): V20 wins F-2 and F-7 outright; on F-1 macro-F1 V2 leads (0.861, V20 0.858) and the nineteen recipes span 0.837 to 0.861. F-7 mean ranking across the six scenes is V12 0.534 / V3 0.524 / V20 0.520, within 0.014 NMI of each other at $Q = 8$: a flat top, not a sharp peak (the ranking inverts to a V20 lead at $Q = 32$).

P1 code base (§III) and run each through the same evaluation framework. The seven extension recipes V13–V15 and V17–V20 add learnt-codebook tokens (V13 vector-quantised VAE, V17 sparse-coding dictionary), signal-transform tokens (V14 continuous wavelet, V15 spectral indices), structure-aware tokens (V18 graph-Laplacian eigenvectors, V19 UMAP coordinates) and a label-aware reweighting (V20 mutual-information-weighted bands). The contribution is not a new method but a comparative result: *which recipe matters for which axis on which scene*, and what that joint structure implies for the choice of canonical recipe in future work. The V16 slot is reserved for a foundation-model wordification (HyperSIGMA / HyperspectralMAE), which is left to a follow-up study because its inference cost would dominate the per-scene computational cost of the labelled-scene panel.

A. Why a sweep, not a winner

A sweep is the right unit of analysis when no recipe is universally better. The nineteen-recipe sweep on the six labelled scenes shows that F-2 has four distinct winners (V1, V3, V12, V20), F-7 has four distinct winners (V3, V8, V12, V20), and the three top recipes across F-7 mean (V12, V3, V20) lie within 0.014 NMI of each other. V1 (current canonical) wins F-2 on two scenes and never wins F-7. V20, the simplest label-aware reweighting of V3, wins both F-2 and F-7 on Indian Pines and is within 0.011 NMI of the F-7 leader on two more scenes (Salinas-A and Kennedy SC). Single-recipe claims invite a counter-example; the sweep avoids the trap and

shows that the wordification landscape has a flat top rather than a sharp peak.

B. Contributions

- 1) A formal definition of the V1–V15, V17–V20 wordification design space, spanning seven axes of freedom: token alphabet, spatial vs. purely spectral aggregation, local vs. global vocabulary, document length regime, signal transform, learnt vs. fixed vocabulary, and label-aware vs. label-blind weighting (§III).
- 2) A per-recipe K-policy that keeps the comparison fair for extreme document-length regimes (V7 with ≤ 6 tokens per document, V9 with 1 token per document, V19 with 3 UMAP-axis tokens) where the canonical $K = \max(4, \min(12, n_{\text{classes}}))$ from P1 over-prescribes K (§IV-A).
- 3) The nineteen-recipe result matrix on the six labelled scenes of the P1 panel for F-1, F-2 and F-7, with unpaired bootstrap intervals on F-1 macro-F1 for V1–V12 and paired (scene, fold) comparisons for all nineteen recipes (§V–X).
- 4) Four *axis-recipe affinities* observed in the sweep: V3 vs. F-7 (three scene wins in the twelve-recipe view, one of them by 0.002; ahead of V1 on four of five scenes), V12 vs. F-2 on dense regimes, V20 vs. F-7 NMI on label-information-bearing scenes, and V18 vs. F-7 on manifold-shaped scenes (Pavia U, Salinas) among the unsupervised structure-aware extension recipes; a fifth,

V7 vs. F-9 on HIDSAG, is stated as a hypothesis that this study does not test (§XII).

- 5) A revised recommendation for the role of V1 going forward: *a reproducibility default, not a universal best* (§XIII).

C. Paper organisation

§III formalises the nineteen recipes. §IV states the per-recipe K-policy and the rest of the experimental setup (carried over verbatim from P1 except where noted). §V reports F-1 (classification), §VI F-2 (coherence), §VII F-7 (topic–label coupling). §VIII narrows F-2 and F-7 to the V13–V20 extension and discusses each extension recipe mechanistically. §X stitches the per-axis results into the joint matrix. §XII discusses the four observed axis–recipe affinities and one hypothesis; §XIII makes the canonical-default recommendation; §XIV notes limitations (the nanopq seeding issue on V11; the sliding-window estimator gap for c_{NPMI} on dense recipes; the deferred F-3 / F-4 / F-5 / F-8 / F-9 / F-10 / F-11 / F-12 sweeps; the V16 foundation-model wordification deferred to a follow-up). Appendix A reproduces the full 19-recipe matrices.

II. RELATED WORK

a) Topic models: Latent Dirichlet allocation [7] and its collapsed-Gibbs [8] and online-variational [9] estimators are the backbone this paper sweeps over; we use the online-VB implementation (1). The nonparametric hierarchical Dirichlet process [10] (2) and the neural topic models ProLDA [11] (3) and the embedded topic model [12] (4) (the latter built on neural variational inference [13] and the variational autoencoder [14]) enter as the alternative backbones of the cross-backbone factorial (§XI). The model run under the ProLDA label is the mixture-decoder form of Srivastava and Sutton’s amortised topic model (their NVLDA): a standard normal prior $\delta_d \sim \mathcal{N}(0, I)$, $\theta_d = \text{softmax}(\delta_d)$ and $w_{dn} \sim \text{Cat}(\sum_k \theta_{dk} \beta_k)$ with $\beta_k = \text{softmax}(\omega_k)$ from free topic–word logits ω_k ; the product-of-experts decoder that defines ProLDA was not run. ETM has the same prior and amortised encoder and the decoder $\beta_k = \text{softmax}(\rho^\top \alpha_k)$ with word embeddings of dimension $L = 64$ and a mixture likelihood, so the two neural backbones differ only in the decoder parameterisation (a free $K \times |V|$ topic–word matrix against a rank- L factorisation). Model selection over K is informed by the estimation-and-selection analysis of [15].

b) LDA and topic models on hyperspectral and spectral data: Probabilistic topic models have been applied to spectral data for subpixel unmixing [16], plant-phenotyping spectroscopy [17], and geometallurgical estimation from hyperspectral core imagery [18], which introduced the band-frequency document recipes that V1–V3 generalise here. The broader hyperspectral classification and unmixing literature [19]–[21] supplies the endmember-extraction machinery behind V8: NFINDR-style simplex-vertex endmembers, with vertex component analysis [22] as the closely-related fast alternative and [23] surveying spectral-variability effects on the abundances V8 quantises.

c) Coherence, reliability, and evaluation: F-2 uses the c_v coherence of [24] (12), which generalises the human-judgement-correlated PMI/NPMI measures of [25] and the document-co-occurrence statistic of [26]. F-7 reports a mutual information normalised by the label entropy $H(Y)$ (13), i.e. the asymmetric uncertainty coefficient $U(Y | \hat{Z}) = I(\hat{Z}; Y)/H(Y)$ (not chance-corrected); the normalisation follows the conventions catalogued by [27]. Because the topic count K varies across recipes and backbones, this uncorrected score carries a mild K -dependent upward bias and is best read within, rather than strictly across, recipes of differing K . F-18 instantiates the multi-seed stability protocol of [28] using Hungarian matching [29], [30] of top- N indicator sets (15). The uniform-quantiser distortion bound of [31] bounds the wordification noise introduced by the Q -level binning in (5). We follow the reproducibility recommendations of [32] and build on the scikit-learn [33], NumPy [34], SciPy [35] and gensim [36] implementations; the mutual-information weighting in V20 (10) uses the `mutual_info_classif` estimator of [33].

III. THE NINETEEN-RECIPE WORDIFICATION DESIGN SPACE

$$\phi_k \sim \text{Dir}(\beta), \quad \theta_d \sim \text{Dir}(\alpha), \quad (1)$$

$$z_{dn} \sim \text{Cat}(\theta_d), \quad w_{dn} \sim \text{Cat}(\phi_{z_{dn}}).$$

$$\beta \sim \text{GEM}(\gamma), \quad \pi_d \sim \text{DP}(\alpha_0, \beta), \quad (2)$$

$$z_{dn} \sim \text{Cat}(\pi_d), \quad w_{dn} \sim \text{Cat}(\phi_{z_{dn}}),$$

with global atom weights $\beta_k = \beta'_k \prod_{l < k} (1 - \beta'_l)$, $\beta'_k \sim \text{Beta}(1, \gamma)$.

$$\delta_d \sim \mathcal{N}(0, I), \quad \theta_d = \text{softmax}(\delta_d), \quad (3)$$

$$\beta_k = \text{softmax}(\omega_k), \quad w_{dn} \sim \text{Cat}\left(\sum_k \theta_{dk} \beta_k\right),$$

where the unnormalised topic–word logits $\omega_k \in \mathbb{R}^{|V|}$ form a free $K \times |V|$ matrix and inference is amortised, $(\mu_d, \Sigma_d) = \text{Enc}_\eta(n_d)$. ProLDA as published mixes the logits instead, $w_{dn} \sim \text{Cat}(\text{softmax}(\sum_k \theta_{dk} \omega_k))$ (a product of experts), with a Laplace approximation to $\text{Dir}(\alpha)$ as the prior; that form was not run.

$$\delta_d \sim \mathcal{N}(0, I), \quad \theta_d = \text{softmax}(\delta_d), \quad (4)$$

$$\beta_k = \text{softmax}(\rho^\top \alpha_k), \quad w_{dn} \sim \text{Cat}\left(\sum_k \theta_{dk} \beta_k\right),$$

with word embeddings $\rho \in \mathbb{R}^{L \times |V|}$ and topic embeddings $\alpha_k \in \mathbb{R}^L$ ($L = 64$), and the same amortised encoder as (3).

$$\tilde{x}_{db} = \frac{x_{db} - \min_b x_{db}}{\max_b x_{db} - \min_b x_{db}}, \quad (5)$$

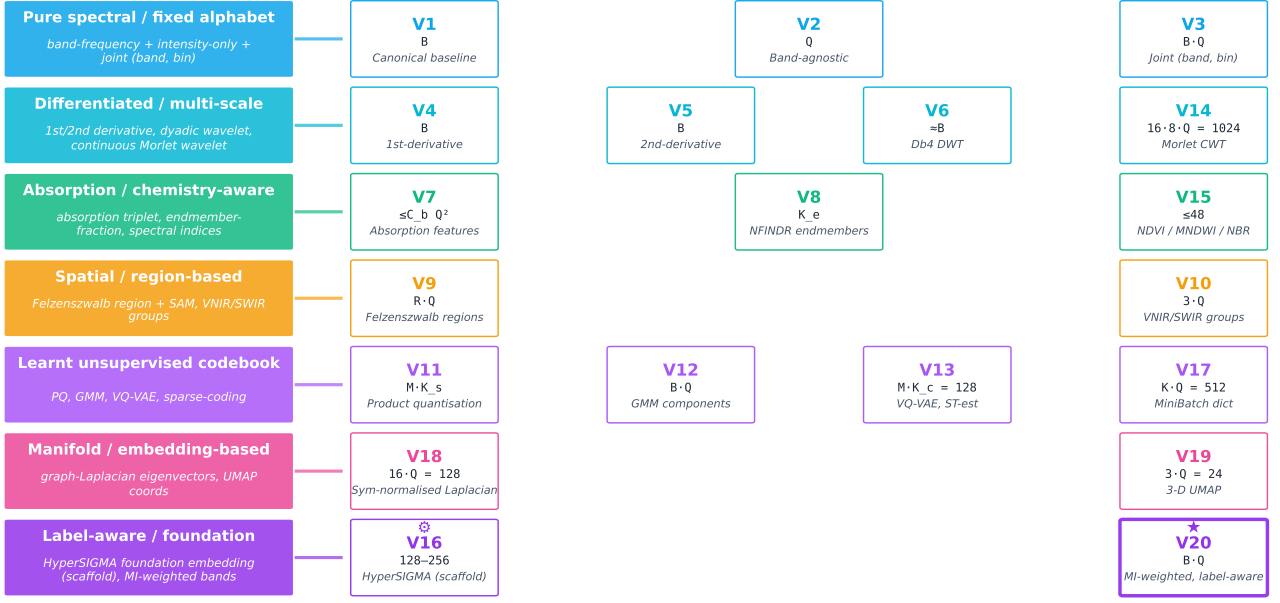
$$q_b(x_d) = \text{clip}(\lfloor \tilde{x}_{db} Q \rfloor, 0, Q - 1).$$

$$n_d^{V1}[b] = q_b(x_d), \quad |V_{V1}| = B. \quad (6)$$

$$n_d^{V3}[(b, q)] = \mathbb{1}[q_b(x_d) = q], \quad |V_{V3}| = BQ. \quad (7)$$

Wordification taxonomy: 20 recipes for LDA-on-HSI

Each group bundles recipes that share token-alphabet semantics. Vocabulary size and core mechanism are noted per recipe.



★ = V20, the only label-aware recipe in the sweep: on Indian Pines it wins F-2 (0.88) and F-7 (0.44); on F-1 (0.858) it is second to V2 (0.861).
⊗ = scaffolded only (V16 reserves the foundation-model slot and is not evaluated).

Fig. 2. The wordification taxonomy. Twenty recipes are grouped into seven families by token-alphabet semantics. V20 (purple, ★) is the only label-aware recipe in the sweep; it wins F-2 and F-7 together on Indian Pines, where it is second to V2 on F-1 (0.858 against 0.861). V16 (HyperSIGMA foundation-model embedding) is scaffolded only and is not evaluated: its pretrained weights are not integrated.

$$a_d = \arg \min_{a \geq 0, \mathbf{1}^\top a = 1} \|x_d - E a\|_2^2, \quad (8)$$

$$n_d^{V8}[m] = \min \left(\left\lfloor Q \frac{a_{dm}}{\max_{m'} a_{dm'}} \right\rfloor, Q - 1 \right),$$

where $E = [e_1, \dots, e_M]$ are the NFINDR endmembers (simplex vertices), x_d is the unnormalised spectrum and $|V_{V8}| = M$. The abundance non-negativity (ANC, $a \geq 0$) and sum-to-one (ASC, $\mathbf{1}^\top a = 1$) constraints make a_d a point on the simplex spanned by the endmembers (fully-constrained least squares, FCLS; implemented as δ -augmented NNLS followed by renormalisation), so the coefficients are proper abundance fractions over the convex hull. The dominant endmember of every pixel emits $Q - 1$ tokens, so the document length grows with Q .

$$g_b(x_d) = \arg \max_{k \in \{0, \dots, Q-1\}} \pi_k \mathcal{N}(x_{db} | \mu_k, \sigma_k^2), \quad (9)$$

$$n_d^{V12}[(b, g)] = \mathbf{1}[g_b(x_d) = g], \quad |V_{V12}| = BQ,$$

where $(\pi_k, \mu_k, \sigma_k^2)_{k < Q}$ is a one-dimensional Gaussian mixture fitted once per scene to the unnormalised band intensities of the sampled pixels (a random subset of 50 000 values), with the components indexed in increasing order of μ_k . Each band emits one token, as in V3, but the bins are shared by all pixels and follow the intensity distribution instead of splitting each spectrum's own range into equal intervals.

$$w_b = \left\lfloor \frac{\text{MI}(x_{:b}; y)}{\max_{b'} \text{MI}(x_{:b'}; y)} Q_{\max} \right\rfloor, \quad (10)$$

$$n_d^{V20}[(b, q)] = w_b \mathbf{1}[q_b(x_d) = q].$$

The token alphabet is V3's, $|V_{V20}| = BQ$ (≈ 1600 at $B = 200, Q = 8$; mean ≈ 1376 across the six labelled scenes since B varies). Bands with $w_b = 0$ emit nothing, so the *effective* (non-empty-column) vocabulary is smaller still (≈ 770 mean).

$$\text{F-1} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \frac{2 P_c R_c}{P_c + R_c}. \quad (11)$$

$$\text{NPMI}(w_i, w_j) = \frac{\log \frac{p(w_i, w_j) + \epsilon}{p(w_i)p(w_j)}}{-\log(p(w_i, w_j) + \epsilon)}, \quad (12)$$

$$c_v = \frac{1}{N} \sum_i \cos(\mathbf{v}_i, \sum_j \mathbf{v}_j),$$

with $\mathbf{v}_i = (\text{NPMI}(w_i, w_\ell))_{\ell=1}^N$ over the top- N topic words.

$$\hat{z}_d = \arg \max_k \theta_{dk},$$

$$\text{F-7} = \frac{I(\hat{Z}; Y)}{H(Y)} = U(Y | \hat{Z}) = 1 - \frac{H(Y | \hat{Z})}{H(Y)}. \quad (13)$$

$$\text{F-14} = \frac{2}{K(K-1)} \sum_{k < k'} \frac{|T_k \cap T_{k'}|}{|T_k \cup T_{k'}|}, \quad T_k = \text{top-}N(\phi_k). \quad (14)$$

$$F-18 = \frac{1}{\binom{S}{2}} \sum_{s < s'} \frac{1}{K} \max_{\sigma \in \mathfrak{S}_K} \sum_k \cos(\mathbf{u}_k^{(s)}, \mathbf{u}_{\sigma(k)}^{(s')}), \quad (15)$$

over S reseeded refits, with $\mathbf{u}_k^{(s)}$ the top- N indicator vector of topic k in run s .

$$\begin{aligned} F-22(d) = \min_{\delta \in \mathbb{Z}[V]} \|\delta\|_1 \text{ s.t.} \\ \arg \max_k \theta_k(n_d + \delta) \neq \arg \max_k \theta_k(n_d), \\ n_d + \delta \geq 0, \end{aligned} \quad (16)$$

reported as the per-scene median, capped at a sentinel $\text{MAX_STEPS} = 50$ when no flip is found.

$$\begin{aligned} \text{aligned}(d) = \mathbb{1} [|\text{top}_N(n_d) \cap \text{top}_N(\phi_{z_d})| \geq \tau \\ \vee \text{top}_3(n_d) \cap \text{top}_N(\phi_{z_d}) \neq \emptyset], \\ \text{decisive}(d) = \mathbb{1} [\text{aligned}(d) = 1 \vee |\text{supp}(n_d)| \geq 3], \\ F-15 = \sum_d \text{aligned}(d) / \sum_d \text{decisive}(d). \end{aligned} \quad (17)$$

$$\begin{aligned} p_{dv} = \frac{n_{dv}}{\sum_{v'} n_{dv'}}, \\ N_{\text{eff}}(d) = \exp(H(p_d)) = \exp\left(-\sum_v p_{dv} \log p_{dv}\right). \end{aligned} \quad (18)$$

N_{eff} counts the tokens that effectively carry the mass of p_d (or of a topic's ϕ_k): it is low when the mass is concentrated on few tokens and at most $|V|$.

$$\mathcal{A}(V_i) = \frac{1}{|\mathcal{B}|} \sum_{m \in \mathcal{B}} \frac{c_v^{(m)}(V_i) - \min_j c_v^{(m)}(V_j)}{\max_j c_v^{(m)}(V_j) - \min_j c_v^{(m)}(V_j)} \in [0, 1], \quad (19)$$

averaged over backbones $\mathcal{B} = \{\text{LDA}, \text{HDP}, \text{ProdLDA}, \text{ETM}\}$; $\mathcal{A} \rightarrow 1$ when a recipe sits near the per-backbone top everywhere.

Equations (1)–(19) are the shared canonical definitions of the paper series: the LDA backbone this paper sweeps over (1), the per-row min–max quantiser (5) that underlies every intensity recipe, the five recipes referenced repeatedly below (V1 (6), V3 (7), V8 (8), V12 (9), V20 (10)), and the six evaluation axes reported here (F-1 (11), F-2 (12), F-7 (13), F-14 (14), F-18 (15), F-22 (16)). The remaining backbones (2)–(4) enter in the cross-backbone factorial of §XI.

We define each recipe as a map $\text{wordify}_V : \mathbb{R}^B \rightarrow \mathbb{N}^{|\mathcal{V}_V|}$ from a normalised pixel spectrum to a non-negative count vector over a recipe-specific vocabulary $\mathcal{V}_V$. Figure 2 groups the nineteen built recipes (V1–V15, V17–V20) into seven semantic families; Table I states each recipe formally and §III-A groups them by seven axes of design freedom.

A. Seven axes of design freedom

- 1) **Token alphabet.** Intensity (V1, V2, V10, V12, V20), local derivative (V4, V5), multi-scale (V6, V14), absorption (V7), mixing fractions (V8, V15), spatial-aware (V9), encoded subspaces (V11, V13, V17, V19), graph-spectral (V18).

- 2) **Spatial vs. spectral.** Pure spectral: V1–V8, V10–V15, V17–V20; spatial-aware via adjacency: V9; spatial-aware via embedding manifold: V18 (kNN graph) and V19 (UMAP).
- 3) **Local vs. global vocabulary.** Local per-band: V1, V3, V4, V5, V12, V20; global band-agnostic: V2, V11, V13, V17, V18, V19; coarse group-level: V10, V8, V14, V15; sparse event-level: V7, V9.
- 4) **Document length** (mean tokens per document on Indian Pines at $Q = 8$). Dense, B or more: V2, V3 and V12 ($B = 200$), V1 (450), V4 (585), V5 (622) and V20 (1187, label-weighted); medium, 14–75: V6 (75), V8 (14), V14 and V18 (16); short, 3–6: V7, V10, V11, V13, V15, V17, V19; a single token: V9.
- 5) **Signal transform.** None (V1–V3, V8, V10, V20), differencing (V4, V5), wavelet (V6, V14), continuum-removed absorption (V7), spectral-index (V15), graph-Laplacian (V18), manifold embedding (V19).
- 6) **Learnt vs. fixed vocabulary.** Fixed (V1–V7, V10, V14, V15); learnt unsupervised (V8 NFINDR, V11 PQ codebook, V12 GMM, V13 VQ-VAE, V17 dictionary, V18 spectral, V19 UMAP); learnt with labels (V20).
- 7) **Label awareness.** Label-blind: V1–V19; label-aware via per-band MI re-weighting: V20.

IV. METHOD

A. Per- V K -policy

The canonical fit defined in P1 uses $K = \max(4, \min(12, n_{\text{classes}}))$. This holds K constant across recipes, which suits recipes with long documents (V1 averages 450 tokens per document on Indian Pines), but over-prescribes K when the document length collapses (V7 at ≤ 6 , V9 at 1, V10 at 3, V11 at 4). We therefore set K from the mean document length $\bar{L}_V$ of each recipe and scene:

$$K_V = \begin{cases} K_{P1}, & \bar{L}_V \geq 8, \\ \text{clip}(\text{round}(\bar{L}_V/2), 3, K_{P1}), & 2.5 \leq \bar{L}_V < 8, \\ 4, & \bar{L}_V < 2.5, \end{cases}$$

where $K_{P1} = \max(4, \min(12, n_{\text{classes}}))$. This keeps V1–V6, V8, V12, V14, V18 and V20 at the P1 K , sets V9 to 4, and reduces V7, V10, V11, V13, V15, V17 and V19 to 3. The one exception is V10 on Pavia U, whose 430–860 nm range covers only one of V10's three band groups: its documents hold a single token and take $K = 4$.

B. Other experimental settings

The topic backbone is online-variational-Bayes LDA (1) throughout (the cross-backbone HDP / ProdLDA / ETM extensions of §XI substitute (2), (3) and (4) respectively). All other settings are inherited verbatim from P1 (the ‘‘Canonical fit’’ subsection of that paper): same stratified-pixel sample, same online-VB hyperparameters, same five-fold StratifiedKFold, same per-fold LDA refit-no-leakage protocol and the same per-fold macro-F1, which §IX summarises with bootstrap intervals.

TABLE I

THE NINETEEN WORDIFICATION RECIPES (V1–V15, V17–V20). VOCABULARY SIZE AND MEAN DOCUMENT LENGTH ARE STATED FOR THE V-SWEEP CANONICAL SETTING (SCHEME = UNIFORM, $Q = 8$, INDIAN PINES, $B = 200$ BANDS); NUMERIC DOCUMENT LENGTHS ARE THE ARCHIVED INDIAN PINES MEANS. V16 IS RESERVED FOR A FOUNDATION-MODEL EMBEDDING WORDIFICATION AND IS NOT BUILT IN THIS STUDY. THE FIVE RECIPES THAT RECUR IN THE RESULTS ARE FORMALISED IN THE CANONICAL EQUATION SET: V1 (6), V3 (7), V8 (8), V12 (9) AND V20 (10); ALL INTENSITY RECIPES SHARE THE QUANTISER $q_b(\cdot)$ OF (5).

ID	Token alphabet	Construction	$ \mathcal{V} $	$\overline{\text{doc len}}$
V1	(band, q-bin)	$\forall b : (b, \text{bin}_Q(x_b))$	B	450
V2	q-bin (band-agnostic)	drop band: $\text{bin}_Q(x_b)$ only	Q	B
V3	joint (band, bin)	$\forall b : (b, \text{bin}_Q(x_b))$ joint vocab	$B \cdot Q$	B
V4	derivative-bin	$(b, \text{bin}_Q(x'_b))$, $x'_b = x_{b+1} - x_b$	B	585
V5	2nd-derivative bin	$(b, \text{bin}_Q(x''_b))$, $x''_b = x_{b-1} - 2x_b + x_{b+1}$	B	622
V6	wavelet coefficient	Db4 level-4 DWT $\{d_k^{(j)}\}$, each binned	$\approx B$	75
V7	absorption triplet	continuum-removed; per feature (centre, depth-bin, area-bin)	$\leq C_b Q^2$	≤ 6
V8	endmember-fraction	NFINDR endmembers; token e_i with count from its NNLS abundance α_i	K_e	14
V9	region + SAM	Felzenszwalb region r , (region, $\text{bin}_Q(\text{SAM})$)	RQ	1
V10	band group	VNIR / SWIR-1 / SWIR-2; $(g, \text{bin}_Q(\bar{x}_g))$	$3Q$	3
V11	product-quantisation	PQ($M = 4$, $K_s = Q$); (m, c_m) per sub-vector	MQ	M
V12	GMM-component token	one 1-D GMM with Q components on the raw band intensities; (b, g_b) where g_b = most probable component	BQ	B
V13	VQ-VAE codebook	PyTorch VQ-VAE, $M = 4$ sub-vectors, $K = 32$ codewords, ST-estimator	$MK = 128$	M
V14	CWT-Morlet	16 log-scales $\times$ 8 position buckets $\times$ Q magnitude bins; top-16 cells by magnitude per pixel	$16 \cdot 8 \cdot Q = 1024$	16
V15	spectral indices	NDVI, MNDWI, NBR, NDSI, EVI, SAVI; per-index q-bin	$N_{\text{idx}} \cdot Q$	N_{idx}
V17	sparse-coding atom	MiniBatch dictionary learning, lasso-LARS, $n_{nz} = 8$; (atom, abundance-bin)	$K_{\text{atoms}}Q = 512$	3.4
V18	graph-Laplacian eigvec	kNN ($K = 10$) cosine affinity, sym-normalised Laplacian; first 16 eigenvectors percentile-binned	$16Q = 128$	16
V19	UMAP coordinate	3D UMAP (n_neighbors=15, min_dist=0.1); per-axis q-bin	$3Q = 24$	3
V20	MI-weighted band	V3 with per-band copies $w_b = \text{round}(\text{MI}_{\text{norm}}(x_b; y) \cdot 8)$	$B \cdot Q$	$\sum_b w_b = 1187$

V. F-1: CLASSIFICATION UNDER V-SWEEP

We follow the F-1 protocol of P1: per-V K-policy, 5-fold StratifiedKFold (`random_state = 42`), per-fold LDA refit with no test-set leakage at the LDA fit step, two methods reported per cell (*raw_logistic* baseline and *topic_routed_soft*, the soft-gated per-topic specialist ensemble from P1). The reported score is the macro-averaged F_1 of (11), computed per fold and averaged.

a) *V20 leakage caveat*: V20 wordification computes per-band mutual information weights using the full labelled stratified sample (`SAMPLES_PER_CLASS = 220`) before the F-1 5-fold split. The resulting `doc_term` matrix therefore encodes test-fold label information into the vocabulary structure. The per-fold LDA refit prevents downstream LDA leakage but does *not* correct the upstream MI weighting. V20's F-1 macro-F1 should be read with this caveat. Its six-scene F-1 (0.917 at $Q = 8$, 0.916 at $Q = 16$ and $Q = 32$) does not exceed that of the label-blind V2 (0.917–0.918) or V8 (0.916–0.917) at

any Q , so the leakage produces no visible F-1 advantage, but V20 is not directly comparable with the label-blind recipes on F-1. A nested protocol (MI weights computed per fold from the training fold only) removes this leakage; its effect on F-1 is not measured here. F-2, F-14, F-18 and F-22 use no labels. F-7 has no held-out fold: it is computed on all sampled pixels, whose labels also set V20's weights, so V20's F-7 measures the label alignment of a representation built with those labels.

Table II gives the mean macro-F1 across folds for *topic_routed_soft* on all 6 labelled scenes.

V12 leads with mean macro-F1 = 0.9216; V1 (current canonical) is sixth at 0.9152. The spread is small (0.0082 across the twelve recipes, 0.0094 across all nineteen): V12 leads on Kennedy SC and Pavia U, V2 on Indian Pines and Salinas, and V8 on Botswana (by 0.0007 over V3). On Salinas-A, the easiest scene, V1 and V6 tie at 0.9970 with four extension recipes, and all nineteen recipes lie within 0.0015.

§IX compares the recipes with unpaired and paired boot-

One spectrum, four token bags: the wordification step of LDA-on-HSI

The identical pixel spectrum (left of each row) is turned into a bag-of-tokens by four recipes, one per design family. Source: stored document of a sampled Soybean-notill pixel of Indian Pines (row 142, column 30; 200 bands, 400-2500 nm).



Fig. 3. From spectrum to LDA tokens. A normalised pixel spectrum $x_d \in \mathbb{R}^B$ is mapped to a doc-term count vector n_d by a recipe-specific wordify_V. The intensity recipes apply the per-row min-max quantiser $q_b(\cdot)$ of (5): V1 emits one band token per band with multiplicity equal to its quantised intensity (6); V3 emits a single joint (band, bin) token per band (7); V20 re-emits each V3 token w_b times, with the per-band weight set by the normalised mutual information of band b with the label (10). The resulting bag of tokens is the document that LDA (1) fits.

TABLE II
F-1 MEAN MACRO-F1 (TOPIC_ROUTED_SOFT, 5 FOLDS) FOR V1–V12; TABLE X (APPENDIX A) GIVES ALL NINETEEN RECIPES. BOLD = BEST PER SCENE AT FULL PRECISION (V1 AND V6 TIE ON SALINAS-A).

Scene	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12
Botswana	0.96	0.97	0.97	0.96	0.96	0.96	0.96	0.97	0.96	0.97	0.96	0.97
Indian Pines	0.84	0.86	0.85	0.84	0.84	0.84	0.84	0.85	0.84	0.84	0.84	0.85
Kennedy SC	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.92	0.91	0.93
Pavia U	0.81	0.80	0.80	0.82	0.81	0.82	0.82	0.81	0.81	0.81	0.83	0.83
Salinas-A	0.997	0.996	0.996	0.996	0.996	0.997	0.996	0.995	0.996	0.996	0.996	0.996
Salinas	0.95	0.96	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95
Mean	0.9152	0.9173	0.9153	0.9151	0.9145	0.9135	0.9143	0.9163	0.9143	0.9134	0.9161	0.9216

strap intervals.

VI. F-2: TOPIC-WORD COHERENCE

F-2 is the c_v coherence of (12) (NPMI sliding window with cosine aggregation, top-10 words per topic). Table III reports the 12×5 matrix for V1–V12 on five labelled scenes; the Botswana row and the V13–V20 recipes are reported in Table XI (Appendix A).

V12 (GMM-token) wins F-2 on three of the five scenes; V1 wins on Kennedy SC and Salinas-A; V3 is competitive everywhere but wins nowhere.

VII. F-7: TOPIC-LABEL COUPLING UNDER V-SWEEP

F-7 is the topic-label normalised mutual information of (13), computed between the argmax topic assignment $\hat{z}_d = \arg \max_k \theta_{dk}$ and the labelled class y_d . Table IV reports

the 12×5 matrix for F-7 NMI between the argmax topic assignment and the labelled class.

Within this 12-recipe view V3 wins F-7 on Kennedy SC, Salinas-A and Indian Pines (the last by 0.002 over V8, 0.430 against 0.428) and V12 wins on Pavia U and Salinas. In the full 19-recipe sweep (Appendix A) V20 wins Indian Pines F-7 at 0.44. *V1 never wins F-7*: on every labelled scene there is at least one wordification that produces more label-aligned topics. This is the strongest single-axis result in the sweep and the key contribution against the V1-canonical assumption of the companion paper.

VIII. F-2 AND F-7 UNDER THE V13–V20 EXTENSION

Table V and Table VI restate F-2 c_v and F-7 normalised MI on a representative subset of recipes (V1, V3, V12 as the V1–V12 reference points, plus the seven extension recipes V13–V15 and V17–V20). The full 19-recipe matrix is reproduced in Appendix A.

TABLE III

F-2 c_v (TOP-10 WORDS PER TOPIC) FOR V1–V12 ON FIVE SCENES. BOLD = BEST PER SCENE. SWEEP RUN 2026-05-26 ON UNIFORM / $Q = 8$.

Scene	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12
Indian Pines	0.32	0.35	0.70	0.32	0.32	0.32	0.27	0.30	0.71	0.27	0.24	0.79
Kennedy SC	0.97	0.76	0.93	0.81	0.93	0.79	0.37	0.52	0.72	0.53	0.25	0.89
Pavia U	0.91	0.43	0.96	0.59	0.40	0.58	0.27	0.21	0.70	0.32	0.22	1.00
Salinas-A	0.96	0.35	0.92	0.32	0.32	0.81	0.32	0.38	0.71	0.23	0.21	0.88
Salinas	0.36	0.35	0.65	0.32	0.32	0.32	0.20	0.21	0.72	0.29	0.22	0.84

TABLE IV

F-7 NORMALISED MUTUAL INFORMATION BETWEEN ARGMAX TOPIC AND LABEL FOR V1–V12 ON FIVE SCENES. BOLD = BEST PER SCENE AT FULL PRECISION. SWEEP RUN 2026-05-26 ON UNIFORM / $Q = 8$.

Scene	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12
Indian Pines	0.34	0.35	0.43	0.25	0.20	0.25	0.16	0.43	0.10	0.08	0.25	0.42
Kennedy SC	0.41	0.40	0.54	0.32	0.25	0.34	0.20	0.17	0.11	0.17	0.22	0.51
Pavia U	0.47	0.38	0.55	0.38	0.14	0.43	0.23	0.54	0.02	0.00	0.21	0.61
Salinas-A	0.62	0.64	0.68	0.37	0.56	0.33	0.16	0.52	0.20	0.28	0.42	0.55
Salinas	0.47	0.54	0.43	0.39	0.44	0.32	0.28	0.56	0.14	0.20	0.33	0.64

TABLE V

F-2 c_v ACROSS THE V1–V20 WORDIFICATION FAMILY, RESTRICTED TO THE V13–V20 EXTENSION PLUS THREE REFERENCE RECIPES (V1, V3, V12). BOLD = BEST IN THIS SUBSET; THE FULL-19 WINNERS ARE CALLED OUT IN THE TEXT. RUN 2026-05-28.

Scene	V1	V3	V12	V13	V14	V15	V17	V18	V19	V20
Indian Pines	0.32	0.70	0.79	0.29	0.46	0.29	0.34	0.58	0.37	0.88
Salinas	0.36	0.65	0.84	0.29	0.53	0.36	0.59	0.51	0.32	0.80
Salinas-A	0.96	0.92	0.88	0.22	0.72	0.39	0.45	0.52	0.38	0.74
Pavia U	0.91	0.96	1.00	0.28	0.76	0.25	0.40	0.66	0.39	0.95
Kennedy SC	0.97	0.93	0.89	0.26	0.76	0.50	0.42	0.63	0.41	0.88
Botswana	0.35	0.90	0.85	0.17	0.52	0.39	0.34	0.50	0.38	0.85
mean	0.64	0.84	0.88	0.25	0.63	0.36	0.42	0.57	0.38	0.85

TABLE VI

F-7 NMI ACROSS THE V1–V20 WORDIFICATION FAMILY, RESTRICTED TO THE V13–V20 EXTENSION PLUS THREE REFERENCE RECIPES (V1, V3, V12). BOLD = BEST IN THIS SUBSET; THE FULL-19 WINNERS ARE CALLED OUT IN THE TEXT. RUN 2026-05-28.

Scene	V1	V3	V12	V13	V14	V15	V17	V18	V19	V20
Indian Pines	0.34	0.43	0.42	0.29	0.31	0.26	0.18	0.27	0.21	0.44
Salinas	0.47	0.43	0.64	0.35	0.51	0.28	0.17	0.50	0.35	0.47
Salinas-A	0.62	0.68	0.55	0.49	0.55	0.51	0.34	0.52	0.33	0.68
Pavia U	0.47	0.55	0.61	0.22	0.49	0.28	0.21	0.52	0.36	0.54
Kennedy SC	0.41	0.54	0.51	0.31	0.43	0.29	0.21	0.28	0.25	0.52
Botswana	0.42	0.52	0.46	0.21	0.44	0.25	0.19	0.47	0.21	0.47
mean	0.45	0.52	0.53	0.31	0.46	0.31	0.22	0.43	0.29	0.52

a) *V20: mutual-information-weighted bands*: V20 is the cheapest answer to “why are all bands weighted equally?” in V3: re-emit each (band, q-bin) token w_b times where $w_b = \text{round}(\text{MI}_{\text{norm}}(x_b; y) \cdot 8)$. Per-band MI is estimated once per scene on the labelled stratified sample (the leakage caveat of §V) with `sklearn.feature_selection.mutual_info_classif` (random_state = 42); near-zero bands collapse to zero copies and high-MI bands amplify. *V20 wins F-2 on Indian Pines (0.88) and F-7 NMI on Indian Pines (0.44)*, places second to V3 on Salinas-A F-7 (0.68 tie), and matches V12 within 0.02 NMI mean. The simplicity of the recipe relative to V12 (no GMM, no out-of-core fit, no per-

scene clustering) makes it the most attractive of the extensions for the “few-knob, label-aware” regime.

b) *V18: graph-Laplacian eigenvector tokens*: V18 builds a $K = 10$ nearest-neighbour cosine-affinity graph over the sampled pixels, computes the symmetric-normalised Laplacian $L = I - D^{-1/2}AD^{-1/2}$, extracts the first 16 eigenvectors via `scipy.sparse.linalg.eigsh` (sigma-magnitude solver), percentile-bins each eigenvector axis into $Q = 8$ bins, and emits one (eigenvector_id, bin) token per axis per pixel. The intuition is that low-frequency Laplacian eigenvectors carry global manifold structure that LDA can latch onto. The mean F-7 NMI of 0.43 places V18 third

among the extension recipes (behind V20 0.52 and V14 0.46) and *V18 produces the most coherent topics on Pavia U among the unsupervised structure-aware extension recipes* (V13, V17, V18, V19; $c_v = 0.66$). We read V18 topics as contiguous manifold regions rather than band-signature templates, a reading these runs do not test.

c) *V14: continuous-wavelet (Morlet) tokens*: V14 emits top-16 magnitude cells from a 16-log-scale, 8-position-bucket CWT-Morlet decomposition of the spectrum. It is the multi-scale analogue of V6 (Db4 DWT). V14 places second among the extension recipes on F-7 mean (0.46) and produces the second-highest c_v on the extension subset on Pavia U and Kennedy SC. V14 outperforms V6, the discrete-wavelet recipe, on four of the six scenes (mean c_v 0.63 against 0.53).

d) *V13 / V17 / V19: underperformers and the cost of expressiveness*: V13 (VQ-VAE codebook), V17 (sparse-coding dictionary), and V19 (UMAP coordinates) all underperform the V1/V3/V12 reference on F-2 and F-7. V13’s mean F-2 of 0.25 is the second lowest in the nineteen-recipe sweep, after V11 (0.23). A plausible reason is that the jointly trained encoder, codebook and decoder (straight-through estimator) produce compact codes suited to reconstruction rather than to the topic-as-mixture-of-tokens assumption of LDA. V17 has the opposite profile: a 512-token vocabulary (64 atoms $\times$ 8 abundance bins) with 3.4 tokens per document on Indian Pines (at most 8 non-zero coefficients per pixel), a sparsity under which LDA finds few coherent topics. V19’s 3-coordinate tokens are too few, and the per-axis percentile binning produces tokens that are abstract (“UMAP-d0 q03”) and cannot align with class-specific spectral signatures. The takeaway is that expressive learnt representations need not produce LDA-friendly token distributions; the recipe must respect the bag-of-tokens assumption.

e) *V15: spectral-index tokens*: V15 emits NDVI/MNDWI/NBR/NDSI/EVI/SAVI per pixel, with each index q-binned. V15 places below V14 on every axis (F-2 mean 0.36 vs 0.63; F-7 mean 0.31 vs 0.46) but is the only recipe whose token names map to a published remote-sensing semantic: in a topic-as-region interpretation, a topic dominated by “NDVI q05–q07” is unambiguously vegetated. V15 should therefore be regarded as a *semantic baseline* rather than as a performance recipe.

IX. BOOTSTRAP COMPARISON ON F-1

We summarise each recipe’s mean macro-F1 with a bootstrap percentile interval ($B = 5000$ resamples of its 30 per-fold scores, `random_state = 42`; Table VII).

The spread between V12 (best) and V6 (worst) is 0.0085, and every recipe’s interval contains every other recipe’s mean. These unpaired intervals are wide because the 30 scores of a recipe mix six scenes whose F-1 runs from 0.80 on Pavia U to 0.997 on Salinas-A, so their overlap does not show that the recipes perform alike. All recipes share the stratified sample and the fold split of each scene, so their fold scores pair. Over the 30 paired (scene, fold) cells, V12 exceeds each of the other eighteen recipes by 0.004 to 0.009 on average (V1: +0.0064, 94% paired bootstrap interval 0.0032 to 0.0100),

TABLE VII
PER-RECIPE BOOTSTRAP MEAN AND 94% PERCENTILE INTERVAL (3% AND 97% QUANTILES OF THE RESAMPLED MEANS) OF THE F-1 MACRO-F1, TOPIC_ROUTED_SOFT, V1–V12.

Recipe	bootstrap mean	94% interval
V12	0.9217	[0.9005, 0.9424]
V2	0.9171	[0.8932, 0.9395]
V11	0.9163	[0.8939, 0.9375]
V8	0.9161	[0.8933, 0.9386]
V3	0.9156	[0.8922, 0.9381]
V1	0.9153	[0.8924, 0.9375]
V4	0.9149	[0.8927, 0.9367]
V5	0.9143	[0.8902, 0.9367]
V9	0.9141	[0.8905, 0.9360]
V7	0.9140	[0.8920, 0.9354]
V10	0.9133	[0.8881, 0.9364]
V6	0.9132	[0.8893, 0.9362]

and the interval excludes zero for every recipe except V2 (+0.0043, -0.0001 to $+0.0095$). The five folds of a scene share training pixels, so these paired intervals are narrower than a resampling of independent cells would give. On F-1 the recipes therefore differ consistently but by less than 0.01. F-2 and F-7 separate them by far more: the spread between the best and worst recipe mean over the five scenes of Tables III and IV is 0.65 on F-2 (V12 against V11) and 0.43 on F-7 (V12 against V9). The paired comparison is computed from the archived per-fold records by `scripts/fl_paired_bootstrap.py` of the manuscript repository https://github.com/fsantibanezleal/CAOS_LDA_HSI_Paper.

X. THE JOINT RESULT MATRIX

Across F-1, F-2 and F-7 on the six labelled scenes, each recipe’s win count is:

TABLE VIII
PER-AXIS WIN COUNT FOR EACH RECIPE OVER THE NINETEEN-RECIPE SWEEP ON THE SIX LABELLED SCENES (F-1: TOPIC_ROUTED_SOFT MACRO-F1). THE SALINAS-A F-1 CELL IS A SIX-WAY TIE AT 0.9970 (V1, V6, V13, V14, V18, V20) AND IS NOT COUNTED. TOTAL = SUM ACROSS F-1, F-2 AND F-7.

Recipe	F-1 wins	F-2 wins	F-7 wins	Total
V12	2	2	2	6
V20	0	1	1	2
V3	0	1	2	3
V1	0	2	0	2
V2	2	0	0	2
V8	1	0	1	2

V12 is the most consistent winner across the three axes, but the nineteen-recipe sweep redistributes the F-2 and F-7 wins of the twelve-recipe view. V20 captures one F-2 win (Indian Pines, 0.88) and one F-7 win (Indian Pines, 0.44); on Indian Pines F-7 four recipes lie within 0.018 NMI (Table XII: V20 0.442, V3 0.430, V8 0.428, V12 0.424). On F-7, V3 keeps its Kennedy SC and Salinas-A wins and loses Indian Pines, its third twelve-recipe win, to V20. V1 (current canonical) wins F-2 on Kennedy SC and Salinas-A, shares the six-way F-1 tie on Salinas-A, and never wins F-7. The spread between V12,

V3 and V20 on F-7 mean is only 0.014 NMI: a flat top, not a sharp peak.

XI. EXTENSION AXES F-13, F-14, F-17, F-18, F-22 AND BACKBONE FACTORIAL

Five further axes extend the framework beyond F-1–F-12, and a backbone factorial repeats F-2 and F-7 under three alternative topic models. Each is described briefly here; the full per-recipe, per-scene tables and the code that builds them are released with the code repository (Data availability). The quantiser sweep $Q \in \{8, 16, 32\}$ of F-2 and F-7 under LDA is shown in Fig. 4 (all nineteen recipes) and Fig. 5 (per-scene F-7 winners); Fig. 6 profiles six recipes over the eight axes, and Fig. 8 gives the rank correlation between the axes.

F-13 SHAP attribution. For every (V, scene) we attribute $\theta_{d,k}$ to recipe-specific tokens using SHAP’s KernelExplainer on a closed-form posterior and report the top-8 tokens per topic by mean absolute SHAP. These attributions are archived (Data availability) and not analysed in this paper.

F-14 repetitiveness. Mean off-diagonal top-10 Jaccard between topics (14). Lower is better. Per-recipe mean across six scenes: V9 = 0.000, V17 = 0.003, V7 = 0.009, V12 = 0.009, V3 = 0.012; V1 = 0.200 (mid-pack); V6 = 0.738, V8 = 0.868 and V2 = 1.000. V2 and V8 meet the vocabulary criterion stated under F-18 below (at most 12 word types), so any two of their top-10 sets share at least 8 tokens and their F-14 values are high by construction; V6’s vocabulary has 131–230 word types, of which 14–58 occur in the documents of a scene.

F-17 cross-scene transfer. Fit ϕ on scene S1, transform S2’s documents, compute F-7 NMI on the target. Only vocab-portable recipes (V2, V10, V11) qualify. V2 transfers best at 0.32 mean NMI; V11 0.19; V10 0.10.

F-18 reliability. Greene et al. (2014) protocol [28]: Hungarian-matched [29] top-10 indicator cosine similarity across reseed pairs (seeds 42–46), as in (15). Per-recipe mean fraction-above-0.7 across the six labelled scenes:

	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12
F-18	0.25	1.00	0.19	0.20	0.15	0.95	0.15	1.00	0.00	0.54	0.28	0.14

Italic marks the recipes that meet the vocabulary criterion used throughout this paper: a vocabulary of at most 12 word types. Any two top-10 sets drawn from at most 12 word types share at least 8 tokens, so every Hungarian-matched pair has cosine at least 0.8 and passes the 0.7 threshold by construction, whatever the fits. V2 (8 word types at $Q = 8$) and V8, whose vocabulary is its set of NFINDR endmembers ($K_e = 12$ word types on four scenes, 9 on Pavia U and 6 on Salinas-A), meet the criterion on every scene, so their F-18 values measure the vocabulary size rather than reseed reliability. V6 (131–230 word types) does not meet it, and its 0.95 is not flagged. The **bold** V12 marks the recipe that leads F-2 and F-7 at $Q = 8$ (its F-1 lead is below 0.01) but loses F-14 to V9, loses F-22 to V20 (V20 26.3 vs V12 24.5), and loses F-18 to every recipe except V9. V12’s flexibility is seed-sensitive.

a) The reliability–informativeness tradeoff: V12, V3, V7, V5, V4 all sit in the 0.14–0.20 range: informative but seed-sensitive. V1 (current canonical) at 0.25 is the most reliable of the per-band recipes (V1, V3, V4, V5, V12), whose

vocabularies have B or $B \cdot Q$ word types; V6 (0.95), V10 (0.54) and V11 (0.28) score higher without meeting the vocabulary criterion. This is the strongest argument for V1 retaining its canonical status *even though V12 leads F-2 and F-7*: V1 is the most reliable of the per-band wordifications.

b) V20 falls on the V12 side of the reliability axis:

The extended F-18 sweep with the label-aware recipe gives V20 a fraction above 0.7 of 0.22 at $Q = 8$ and at $Q = 16$ (mean matched cosine 0.451 and 0.450) across the six labelled scenes, inside the 0.14–0.25 band of the per-band recipes V12, V3, V4 and V1, despite V20’s strong F-2 / F-7 / F-22 performance. A possible reason, which these runs do not test, is that the down-weighted low-MI bands leave parts of the topic–word matrix flat, so different LDA seeds refine the high-MI bands differently. **V8 (NFINDR endmember-fraction)** reaches a mean matched cosine of 0.957 ($Q = 8$) and 0.962 ($Q = 16$), but it meets the vocabulary criterion: its top-10 sets cover all or nearly all of its 6–12 endmember tokens, so every matched pair has cosine at least 0.8 whatever the fits, and these values do not measure reseed reliability. V8’s cross-backbone portability (top-3 mean F-7 NMI across LDA / HDP / ProdLDA / ETM, §XI-A) does not rest on F-18.

F-22 counterfactual. The minimum- L_1 token perturbation that flips $\arg \max_k \theta_k(n_d)$, as defined in (16); we solve it by coordinate-descent search capped at MAX_STEPS = 50 token-flips. Cells where every sampled document fails to flip persist with the sentinel value 50 rather than being dropped. Higher = more robust topic boundary. Per-recipe mean of the per-scene medians across the 19-recipe sweep:

recipe	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10
L_1	6.1	7.8	23.5	2.5	2.5	1.9	5.2	3.6	1.0	1.2
recipe	V11	V12	V13	V14	V15	V17	V18	V19	V20	
L_1	3.3	24.5	2.7	7.7	4.0	2.6	3.7	2.1	26.3	

V20 has the highest F-22 in the sweep ($L_1 = 26.3$), surpassing V12 (24.5), the most robust of V1–V12, and the joint (band, q-bin) vocabulary V3 (23.5). On three scenes (Salinas-A, Pavia U, Botswana) every sampled document under V20 failed to flip within 50 perturbations; the sentinel value 50 caps that observation. F-22 counts unit token changes, so it grows with the document length: a V20 document holds 1187 tokens on Indian Pines against 200 for V3 and V12 (each band repeated w_b times, 5.9 on average), and part of V20’s lead may reflect that scale rather than a more robust topic boundary. V9 (one token per document) flips at the first perturbation.

c) F-22 Q-trajectory: Re-running the four top contenders at $Q \in \{8, 16, 32\}$ reveals a non-monotonic peak for V20 at $Q = 16$ ($L_1 = 41.83$, +59% over $Q = 8$, with five of the six scenes at the sentinel 50), falling back to 26.17 at $Q = 32$. V12 leads at $Q = 32$ ($L_1 = 30.00$, after 24.50 at $Q = 8$ and 20.67 at $Q = 16$). V8 is the only top contender whose L_1 rises monotonically with Q (3.58, 9.58 and 20.25 at $Q = 8, 16$ and 32), roughly in proportion to Q : its counts are NNLS abundances quantised to Q levels, so the rise may reflect the count scale rather than preserved discriminative structure.

Backbone factorial: HDP. The first alternative backbone of the factorial covers the same 114 cells (nineteen recipes on the

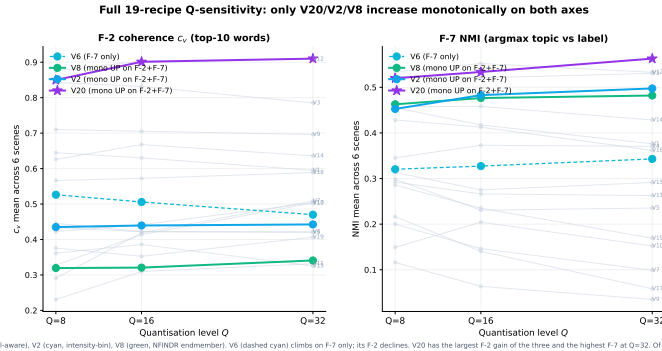


Fig. 4. Full 19-recipe Q-sensitivity under LDA. Both F-2 c_v and F-7 NMI plotted across $Q \in \{8, 16, 32\}$ for every recipe (V13 is excluded because it is insensitive to Q). **Three recipes climb monotonically on both F-2 and F-7:** V20 (purple, $\star$, label-aware), V2 (cyan, intensity-bin), V8 (green, NFINDR endmember). V6 (dashed cyan) is monotonic on F-7 only. The remaining 14 recipes are grey: on F-7, seven decline monotonically (V1, V7, V9, V11, V17, V18, V19), four peak at $Q = 16$ (V4, V10, V12, V14) and three dip at $Q = 16$ (V3, V5, V15). Of the three rising recipes, V20 has the largest F-2 gain (+0.060); its F-7 gain is +0.043 (V2: +0.045). V20 has the highest absolute F-7 at $Q = 32$ (0.563); on F-2 it ties V12 (0.910 vs 0.909). V20 is the only *label-aware* recipe in the monotonic-improvement family.

Per-scene F-7 NMI winner across Q-levels (LDA backbone)

Colour = recipe family from the wordification taxonomy. V20 (purple) emerges at $Q=32$ on Salinas-A, Pavia U, KSC, Botswana.

	Indian Pines	Salinas	Salinas-A	Pavia U	KSC	Botswana
Q=8	V20 NMI = 0.442	V12 NMI = 0.643	V3 NMI = 0.676	V12 NMI = 0.607	V3 NMI = 0.535	V8 NMI = 0.559
Q=16	V20 NMI = 0.473	V12 NMI = 0.677	V2 NMI = 0.695	V3 NMI = 0.555	V3 NMI = 0.524	V8 NMI = 0.556
Q=32	V2 NMI = 0.458	V12 NMI = 0.632	V20 NMI = 0.722	V20 NMI = 0.580	V20 NMI = 0.497	V20 NMI = 0.554

Pure spectral (V1-V3)	Absorption / chemistry (V7, V8, V15)	Manifold (V18, V19)
Differentiated / wavelet (V4-V6, V14)	Learnt codebook (V11-V13, V17)	Label-aware (V20)

Fig. 5. Per-scene F-7 NMI winner under LDA at each quantisation level. V20 dominance scales with Q : at $Q = 8$ V20 wins only on Indian Pines (1/6 scenes), at $Q = 16$ V20 still wins only Indian Pines (1/6), and at $Q = 32$ V20 wins four of six scenes (Salinas-A, Pavia U, KSC, Botswana). The jump from 1/6 to 4/6 between $Q=16$ and $Q=32$ is the sharpest single Q -step gain in the sweep. The $Q=8$ row gives V12 two wins, V3 two, V20 one and V8 one; V12 falls to one win at $Q = 16$ and at $Q = 32$, and V3 keeps two at $Q = 16$ and has none at $Q = 32$. The rise is consistent with V20's per-band MI re-weighting gaining from finer intensity bins.

six labelled scenes, Fig. 7), with gensim's Hierarchical Dirichlet Process in place of fixed-K LDA. For five representative recipes, the per-recipe ranking on F-2 c_v (six-scene means, as in Fig. 7) *inverts*:

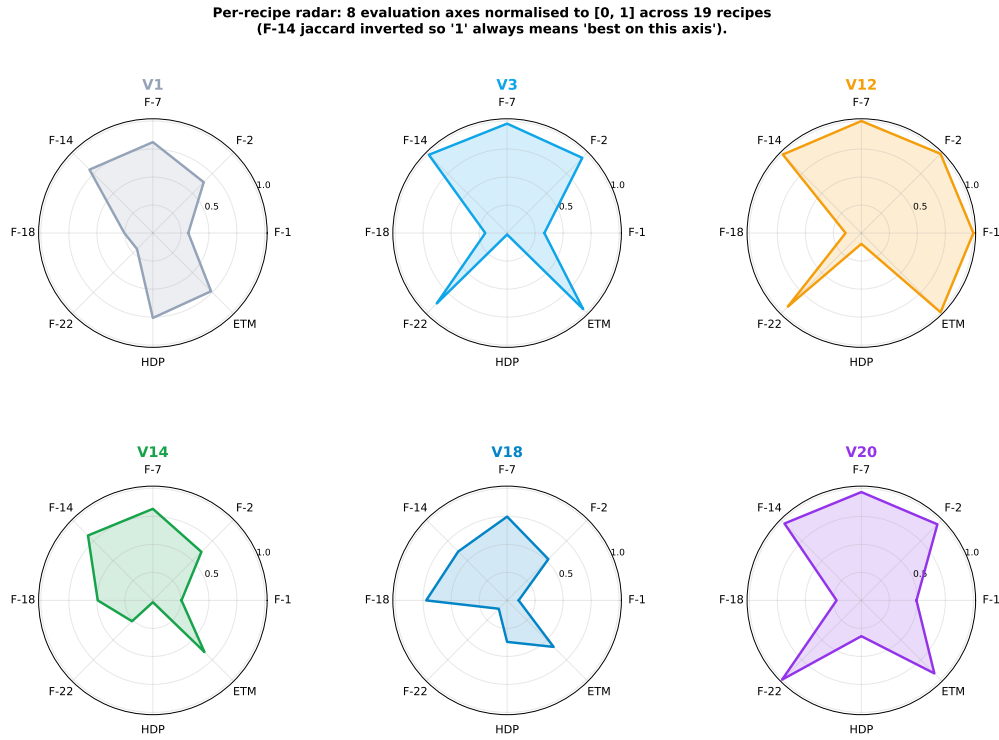
LDA backbone	c_v mean	HDP backbone	c_v mean
V12	0.876	V7	0.615
V3	0.843	V1	0.540
V1	0.645	V11	0.501
V7	0.291	V12	0.337
V11	0.231	V3	0.311

Under LDA the dense GMM-token vocabulary of V12

has the most coherent topics; under HDP, V7 (at most six absorption-feature tokens per pixel) leads. The HDP result is consistent with the truncated Dirichlet-process prior favouring recipes whose few tokens each carry semantic weight, a hypothesis these runs do not test. **This inversion is the cleanest evidence that the wordification ranking depends on the backbone, which motivates the factorial study.**

A. F-7 NMI across the four backbones

The backbone factorial above reports F-2 c_v only; Table IX adds F-7 NMI under each non-LDA backbone.



V20 (purple) and V12 (orange) cover the most surface area: they are top-tier on coherence, counterfactual robustness and the ETM backbone. V18 (cyan) tops the F-18 reliability axis. V1 (grey) is mid-pack on most axes. V14 (green) is a balanced multi-scale alternative.

Fig. 6. Per-recipe radar across the eight evaluation axes, normalised to [0, 1] across the 19-recipe set. F-14 (jaccard) is sign-inverted so “1” always means “best on this axis”. V20 (purple) and V12 (orange) cover the most surface area; V18 (cyan) has the highest F-18 of the six recipes shown; V1 (grey) is mid-pack on most axes.

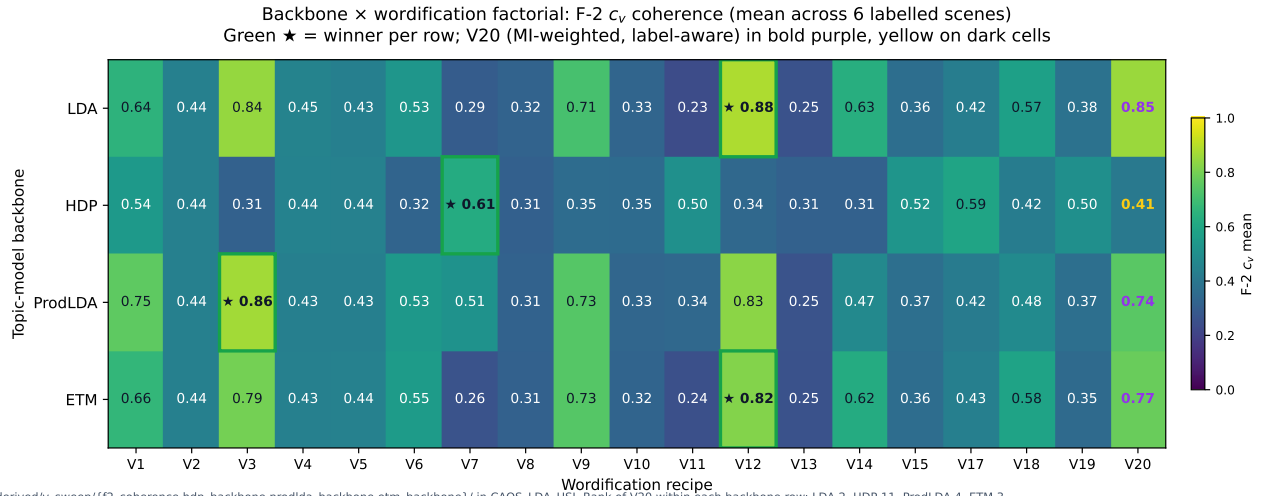


Fig. 7. Backbone × wordification factorial. Each cell shows the mean F-2 c_v across the six labelled scenes for one (backbone, recipe) pair; one row per backbone, recipes in index order. Green border + ★ = per-backbone winner; V20 (the label-aware recipe) is labelled in bold purple (yellow on dark cells), and the footer gives its rank within each row (LDA 2, HDP 11, ProdLDA 4, ETM 3; under HDP, V20 is tenth without the V15 cell noted at the end of this caption). No recipe dominates every backbone: LDA and ETM pick V12, with V3 and V20 close behind; HDP picks V7 (sparse absorption tokens); ProdLDA picks V3. The HDP cell for V15 (0.52) was computed on the stored V15 wordification that holds the $Q = 32$ vocabulary (192 tokens; 96 on Pavia U); every other V15 value at $Q = 8$ in this paper uses the $Q = 8$ vocabulary (48 tokens; 24 on Pavia U).

a) Degenerate cross-backbone fits: The 4-backbone × 19-recipe × 6-scene $Q=8$ factorial contains a non-trivial number of degenerate cells (F-7 below 0.01, an argmax topic that carries no label information): LDA 1/114, ETM 1/114, but

HDP 25/114 (22%) and **ProdLDA 41/114 (36%)**. For HDP, 22 of the 25 are on V1, V4, V5 and V6, the recipes for which HDP keeps between 1.2 and 2.2 effective topics (six-scene means of the F-2 runs); across the nineteen recipes HDP’s F-

TABLE IX

F-7 NMI MEAN ACROSS THE SIX LABELLED SCENES, TOP-12 RECIPES $\times$ FOUR TOPIC-MODEL BACKBONES (FULL SWEEP, 456/456 CELLS). BOLD = PER-BACKBONE WINNER. RECIPES SORTED BY 4-BACKBONE MEAN. LDA USES THE PER-RECIPE K OF §IV-A; THE F-7 RUNS OF PRODLDA AND ETM USE $K = K_{P1}$ FOR EVERY RECIPE, AND HDP TRUNCATES AT 20 TOPICS.

Recipe	LDA	HDP	ProdLDA	ETM	Mean
V8 (NFINDR)	0.463	0.451	0.328	0.482	0.431
V20 (MI-weighted)	0.520	0.356	0.221	0.490	0.397
V2 (intensity-bin)	0.453	0.347	0.324	0.456	0.395
V11 (product quantisation)	0.292	0.571	0.088	0.530	0.370
V12 (GMM-token)	0.534	0.220	0.238	0.488	0.370
V3 (joint band-bin)	0.524	0.200	0.169	0.443	0.334
V19 (UMAP coord)	0.286	0.530	0.051	0.394	0.315
V15 (spectral indices)	0.313	0.438	0.058	0.449	0.315
V13 (VQ-VAE)	0.311	0.399	0.113	0.406	0.307
V14 (CWT-Morlet)	0.457	0.220	0.114	0.430	0.306
V18 (graph-Laplacian)	0.428	0.198	0.080	0.330	0.259
V1 (band-frequency)	0.455	0.081	0.091	0.381	0.252

7 follows the number of topics it keeps (Spearman $\rho = 0.59$). The cross-backbone mean reported in the table above averages these zeros together with the other fits.

Excluding zero cells from the per-backbone means yields V8 0.431 (unchanged, since V8 has no degenerate cells), V20 0.408 (+0.011, one ProdLDA zero on Indian Pines), V11 0.381 (+0.011), V12 0.393 (+0.023). V8 retains cross-backbone leadership under either treatment of zeros. The F-7 value of every (backbone, recipe, scene) cell is archived in the code repository.

V8 (NFINDR endmember-fraction) is the most cross-backbone-consistent recipe at mean F-7 NMI = 0.431; it lands top-5 under every backbone, winning outright only under ProdLDA, and shares with V2 the smallest standard deviation across the four backbones ($\sigma = 0.06$ for both). V20 is the second-most consistent at 0.397, retaining LDA strength (0.520) and tying V12 on ETM (0.490 vs 0.488; V11 leads ETM at 0.530). V11 (product quantisation, vocab $MK_s = 32$) wins HDP and ETM outright but collapses under ProdLDA (0.088) and ranks low under LDA (0.292). F-7 is bounded by $\log K/H(Y)$, and under LDA V11’s four-token documents take $K = 3$, which caps its six-scene mean at 0.46; its HDP and ETM means (0.571 and 0.530) exceed that cap, so this apparent backbone-specialist profile reflects the topic count at least as much as the backbone. The same holds for V19 under HDP (0.530). V20’s interpretability advantage relative to V8 and V11 is that it is the only *label-aware* recipe in the consistently-strong family; V8 and V11 are label-unaware compressions of the spectrum. The cross-backbone story is therefore not “one recipe wins everywhere” but “three recipes lead under different backbones, all above the canonical V1 baseline”.

b) HDP backbone at $Q = 16$: Re-running HDP F-7 NMI for the top-3 contenders at the finer quantisation level reveals that V8’s cross-backbone dominance is preserved under quantisation refinement: HDP $Q = 16$ gives V8 0.454, V20 0.403, V2 0.249, mirroring the $Q = 8$ HDP ranking (V8 = 0.451, V20 = 0.356, V2 0.347). V8’s endmember-fraction vocabulary is robust both across topic-model backbones and across quantisation levels. V20’s gain from finer

Cross-axis Spearman rank correlation across 114 (recipe, scene) cells
F-14 jaccard is sign-inverted so all axes are direction-aligned (higher = better).

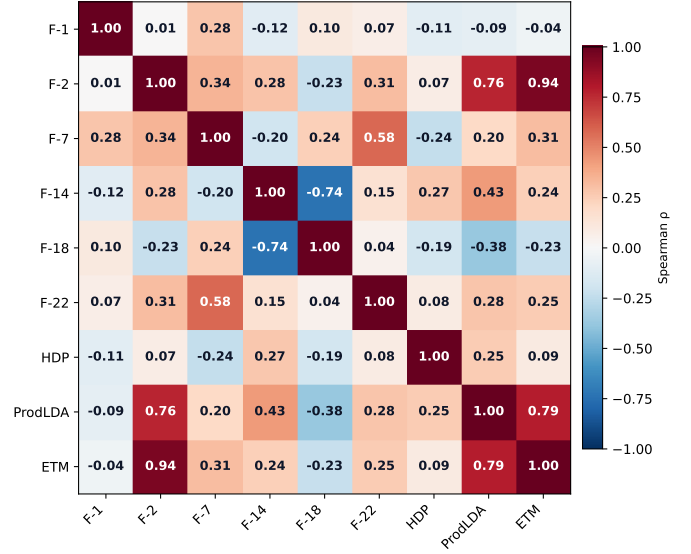


Fig. 8. Spearman rank correlation across the 114 (recipe, scene) cells; HDP, ProdLDA and ETM denote F-2 c_v under that backbone. F-14 inverted so all axes read “higher = better”. F-2 / ETM and F-2 / ProdLDA cluster tightly ($\rho \geq 0.76$); F-14 anti-correlates with F-18 ($\rho = -0.74$, the vocabulary-size confounder); F-22 / F-7 correlate at $\rho = 0.58$ (robust topics tend to be label-aligned). The HDP column includes the V15 cells computed on the $Q = 32$ vocabulary (Fig. 7).

Q is not LDA-specific: from $Q = 8$ to $Q = 16$ it rises under HDP (0.356 to 0.403), ETM (0.490 to 0.525) and LDA (0.520 to 0.534) and stays level under ProdLDA (0.221 to 0.223).

c) ETM backbone at $Q = 16$: Under the embedded topic model backbone, the $Q = 16$ ranking inverts: V20 0.525, V2 0.492, V8 0.473. The MI-weighted recipe now leads.

d) ProdLDA backbone at $Q = 16$: The amortized variational backbone reverses again: V8 0.359, V2 0.335, V20 0.223. The ProdLDA-labelled model shares ETM’s prior and encoder (§II), so V20’s fall from 0.525 under ETM to 0.223 here comes from the decoder parameterisation or from optimisation, which these runs do not separate.

e) Cross-backbone mean F-7 at $Q = 16$: Averaging across the four backbones at $Q = 16$:

Recipe	LDA	HDP	ProdLDA	ETM	Mean
V8	0.476	0.454	0.359	0.473	0.441
V20	0.534	0.403	0.223	0.525	0.421
V2	0.483	0.249	0.335	0.492	0.390

V8 retains its cross-backbone leadership at $Q = 16$ (0.441 mean, ahead of V20’s 0.421), with a backbone split among these three recipes: V8 leads under HDP and ProdLDA, V20 under LDA and ETM. The split does not follow amortized inference: ProdLDA and ETM are amortized, whereas LDA (online variational Bayes) and HDP are not. The backbone-recipe interaction is therefore non-trivial across Q : the backbone the practitioner commits to determines whether label-aware weighting (V20) or geometric endmember decomposition (V8) carries more signal.

f) V8 stable on F-18 across Q : Extending F-18 to $Q=16/32$ on the V20 / V8 / V2 trio yields mean matched

cosine across the six labelled scenes (italic: values under the vocabulary criterion):

Recipe	Q=8	Q=16	Q=32
V8	0.957	0.962	0.959
V2	1.000	0.875	0.781
V20	0.451	0.450	0.443

V8 stays between 0.957 and 0.962 on the F-18 mean matched cosine across the three Q levels. Its vocabulary does not grow with Q (the same 6–12 endmember tokens at $Q = 8, 16$ and 32), so it meets the vocabulary criterion at every Q : every matched pair has cosine at least 0.8 whatever the fits, and these values do not measure reseed behaviour. V20 stays near 0.45 at every Q , inside the range of the per-band recipes at $Q = 8$ (0.35 for V12 to 0.58 for V1). V2 falls with Q (1.000, 0.875 and 0.781 at $Q = 8, 16$ and 32) as its vocabulary grows from 8 to 32 word types and leaves the criterion, confirming that V2's $Q = 8$ value of 1.0 is the vocabulary-size artefact flagged above.

The recommendation therefore rests on cross-backbone F-7 portability (Table IX and the $Q = 16$ table above), not on F-18: **V8 is the recipe of choice when the topic-model backbone is uncertain**; V20 is the recipe of choice when LDA is fixed and one can afford $Q \geq 16$ and accept the seed-sensitivity tradeoff.

XII. FOUR AXIS–RECIPE AFFINITIES AND ONE HYPOTHESIS

The nineteen-recipe sweep exposes four axis–recipe affinities (§XII-A–XII-D); a fifth is a hypothesis that this study does not test (§XII-E).

A. V3 vs. F-7 topic–label coupling

In the twelve-recipe view of Table IV, V3 (the joint (band, bin) vocabulary) wins F-7 NMI on Kennedy SC (0.54), Salinas-A (0.68) and Indian Pines (0.430 against V8's 0.428), while V12 wins on Pavia U (0.61) and Salinas (0.64). V3 exceeds V1 by 0.06–0.13 NMI on four of the five scenes and trails it on Salinas (0.43 vs. 0.47). A likely mechanism: V3 carries $B \times Q$ word types (1600 at $B = 200, Q = 8$) versus V1's B , and the larger vocabulary lets a single topic concentrate on intensity-specific band signatures that map onto land-cover classes. On F-1 macro-F1 the recipes differ by less than 0.01 (§IX).

B. V12 vs. F-2 coherence on dense regimes

In the twelve-recipe view of Table III, V12 (Gaussian-mixture-component tokens) wins F-2 c_v on Indian Pines ($c_v = 0.79$ vs. V1's 0.32), a scene that V20 (0.88) takes in the nineteen-recipe sweep (Table XI), and it is competitive on the other dense scenes. A likely reason is that the mixture components follow the intensity distribution of the scene instead of splitting each spectrum's range into equal bins (9); this study does not separate that effect from V12's other difference from V3, the absence of per-spectrum normalisation.

C. V20 vs. F-7 label-information-rich scenes

V20 (MI-weighted bands) wins F-7 on Indian Pines (0.44, the most class-imbalanced scene), ties V3 on Salinas-A (0.68, the easiest scene by F-1), and matches V12 to within 0.02 NMI mean across the rest of the panel. On Indian Pines its weights are broad rather than concentrated: 169 of the 200 bands receive at least 5 of the 8 possible copies and 22 receive 3 or fewer, one of them none, so V20 differs from V3 mainly by down-weighting a minority of low-MI bands. Its weights come from the labels that F-7 is scored against (§V). The recipe is parameterised by a single integer ($\text{MAX_COPIES} = 8$) and adds no per-fit cost, because the MI estimate is computed once per scene.

D. V18 vs. F-7 on manifold-shaped scenes

Among the unsupervised structure-aware extension recipes (V13, V17, V18, V19), V18 (graph-Laplacian eigenvectors) has the highest F-7 NMI on Pavia U (0.52) and Salinas (0.50, Table VI), and it places third on F-7 mean among the extension recipes (0.43 NMI). Against all nineteen recipes it leads neither scene. The intuition: scenes whose labelled classes correspond to connected regions in the pixel-similarity graph (urban land-cover types in Pavia, agricultural fields in Salinas) yield informative Laplacian eigenvectors whose low frequencies group those regions. Bin-tokens over those eigenvectors then carry segmentation semantics that LDA can fit topics around. The result is a recipe that explicitly exposes the manifold structure to the topic model.

E. V7 vs. F-9 HIDSAG cross-preprocessing (hypothesis)

This fifth affinity is a hypothesis that the present study does not test: F-9 is not swept (§XIV). V7 is the most chemistry-faithful recipe (centroid, depth, area triplets), and we expect it to dominate on the HIDSAG mineralogical subsets, where USGS-library matching is the gold standard.

XIII. RECOMMENDATION: V1 AS CANONICAL DEFAULT, NOT UNIVERSAL BEST

The sweep supports a sharper statement about V1's role: V1 should remain canonical as a *reproducibility default* (deterministic, dense, well-understood, and the band–frequency representation of the geometallurgical study [18] that V1–V3 generalise) but should not be claimed as a universal best. P1's framework results, computed on V1, are stable benchmarks; a future study comparing V1 against, say, V3 or V12 on F-7 would need to refit. The proposed convention is to report every framework result against V1 alongside the per-axis best recipe.

XIV. LIMITATIONS AND FUTURE WORK

- **V11 reproducibility:** the implementation does not pin a random seed for the nanopq codebook fit, so the V11 results reported here come from a single unseeded codebook fit and may change when the seed is pinned and the cells are recomputed.

- **CNPMI on dense recipes:** V1 (and any other recipe with mean doc length close to $|\mathcal{V}|$) produces a degenerate pixel-as-document NPMI because nearly every word fires in every document. A sliding-window estimator would yield comparable numbers; we leave that to a follow-up.
- **F-3 / F-4 / F-5 / F-8 / F-9 / F-10 / F-11 / F-12 not swept:** the sweep reports F-1, F-2 and F-7, together with the extension axes of §XI. The remaining framework axes are left to future work, either as an extension of this study or as a separate one.
- **V16 foundation-model embeddings not built:** V16 is reserved for a foundation-model wordification (HyperSIGMA, HyperspectralMAE, or SatMAE) that embeds each pixel into a learnt latent space and quantises that into LDA tokens. Building V16 requires GPU inference at scene scale, which exceeds the single-node computational resources used in this study. We expect V16 to set the upper bound for F-7 NMI on the panel; its slot in the recipe index is reserved for that follow-up.
- **Full backbone factorial at $Q = 8$ only:** at $Q = 8$ the factorial covers all nineteen recipes on the six labelled scenes under the four backbones, for F-2 (456 cells, Fig. 7) and F-7 (456 cells, Table IX). At $Q = 16$ the HDP, ProdLDA and ETM backbones cover only V2, V8 and V20, and $Q = 32$ is run under LDA only.

XV. CONCLUSION

A nineteen-recipe sweep of the wordification step, evaluated on the F-1, F-2 and F-7 axes of the twelve-axis framework of the companion paper and on five extension axes, shows that *no single recipe is universally best*. On classification (F-1, (11)) the recipes differ consistently but by less than 0.01 macro-F1 (V12 leads, §IX); the discriminating axes are coherence (F-2, (12)) and topic-label coupling (F-7, (13)), where the spread between the best and worst recipe mean reaches 0.65 and 0.43 respectively (Tables III and IV). The same three recipes, V12 (9) (Gaussian-mixture tokens), V3 (7) (joint band-bin vocabulary) and V20 (10) (mutual-information-weighted bands), have the three highest six-scene means on both F-2 and F-7, and on F-7 they lie within 0.014 NMI of each other, a flat design-space ceiling rather than a sharp peak. V20, the simplest label-aware reweighting of V3 (7), wins F-2 and F-7 together on Indian Pines (its weights use the labels that F-7 is scored against) and has the highest F-22 of the sweep ((16), $L_1 = 26.3$, a count that grows with its longer documents); its F-7 lead grows with quantiser resolution, reaching the highest absolute F-7 at $Q = 32$. The cross-backbone factorial identifies V8 (8) (NFINDR endmember-fraction) as the most backbone-portable recipe; its F-18 values ((15), ≥ 0.95) are bounded below by 0.8 through its 6–12-token endmember vocabulary and do not measure reseed reliability. The practical recommendation is twofold: retain V1 as a *reproducibility default* rather than a universal best, and pick the recipe to the axis: V20 when LDA is fixed and finer quantisation is affordable, V8 when the backbone is uncertain. Closing the remaining framework axes (F-3–F-12) over the full nineteen-recipe family, and building the deferred foundation-model wordification V16, are the natural next steps.

DATA AVAILABILITY

Per-cell results (one JSON record per scene, recipe and quantisation level) are archived in the code repository https://github.com/fsantibanezleal/CAOS_LDA_HSI under `data/derived/v_sweep/: f1_per_fold/` (with the unpaired bootstrap summary in `f1_bootstrap_posterior.json`; the paired comparison of §IX and Table X are computed from these records by `scripts/f1_paired_bootstrap.py` of the manuscript repository), `f2_coherence/`, `f7_topic_to_label/`, `f13_shap/`, `f14_repetitiveness/`, `f17_cross_scene/`, `f18_reliability/` and `f22_counterfactual/` for the LDA axes; `hdp_backbone/`, `prodllda_backbone/` and `etm_backbone/` (F-2) and `backbones_f7/` (F-7) for the backbone factorial. The same folder holds runs on the five HIDSAG subsets [37] (`hidsag/: F-2, F-14` and `topic-to-owner` mutual information for fifteen recipes), which this paper does not analyse.

APPENDIX A

FULL NINETEEN-RECIPE F-1, F-2 AND F-7 MATRICES

Tables X, XI and XII report F-1 macro-F1, F-2 c_v and F-7 NMI for all nineteen recipes on the six labelled scenes.

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TABLE X

F-1 MEAN MACRO-F1 (TOPIC_ROUTED_SOFT, 5 FOLDS) ACROSS THE FULL V1–V15, V17–V20 SWEEP ON THE LABELLED-SCENE PANEL. BOLD = BEST PER SCENE AT FULL PRECISION, EXACT TIES INCLUDED (V8 LEADS V3 ON BOTSWANA BY 0.0007).

Scene	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14	V15	V17	V18	V19	V20
Botswana	0.963	0.965	0.966	0.964	0.963	0.959	0.963	0.966	0.962	0.965	0.963	0.965	0.964	0.964	0.962	0.963	0.965	0.965	0.964
Indian Pines	0.842	0.861	0.852	0.844	0.841	0.839	0.840	0.850	0.845	0.842	0.845	0.853	0.839	0.847	0.844	0.843	0.837	0.840	0.858
Kennedy SC	0.923	0.921	0.923	0.919	0.920	0.919	0.916	0.920	0.919	0.916	0.915	0.930	0.919	0.917	0.919	0.920	0.919	0.918	0.921
Pavia U	0.815	0.804	0.805	0.818	0.815	0.815	0.821	0.814	0.813	0.808	0.828	0.834	0.805	0.811	0.814	0.803	0.810	0.805	0.811
Salinas-A	0.997	0.996	0.996	0.996	0.996	0.997	0.996	0.995	0.996	0.996	0.996	0.996	0.997	0.997	0.996	0.996	0.997	0.996	0.997
Salinas	0.951	0.956	0.950	0.950	0.952	0.951	0.948	0.952	0.951	0.952	0.950	0.952	0.952	0.951	0.954	0.950	0.951	0.949	0.950
mean	0.915	0.917	0.915	0.915	0.914	0.913	0.914	0.916	0.914	0.913	0.916	0.922	0.912	0.915	0.915	0.913	0.913	0.912	0.917

TABLE XI

F-2 c_v ACROSS THE FULL V1–V15, V17–V20 SWEEP ON THE LABELLED-SCENE PANEL. BOLD = BEST PER SCENE ACROSS ALL 19 RECIPES.

Scene	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14	V15	V17	V18	V19	V20
Indian Pines	0.32	0.35	0.70	0.32	0.32	0.32	0.27	0.30	0.71	0.27	0.24	0.79	0.29	0.46	0.29	0.34	0.58	0.37	0.88
Salinas	0.36	0.35	0.65	0.32	0.32	0.32	0.20	0.21	0.72	0.29	0.22	0.84	0.29	0.53	0.36	0.59	0.51	0.32	0.80
Salinas-A	0.96	0.35	0.92	0.32	0.32	0.81	0.32	0.38	0.71	0.23	0.21	0.88	0.22	0.72	0.39	0.45	0.52	0.38	0.74
Pavia U	0.91	0.43	0.96	0.59	0.40	0.58	0.27	0.21	0.70	0.32	0.22	1.00	0.28	0.76	0.25	0.40	0.66	0.39	0.95
Kennedy SC	0.97	0.76	0.93	0.81	0.93	0.79	0.37	0.52	0.72	0.53	0.25	0.89	0.26	0.76	0.50	0.42	0.63	0.41	0.88
Botswana	0.35	0.35	0.90	0.32	0.32	0.34	0.33	0.29	0.70	0.33	0.26	0.85	0.17	0.52	0.39	0.34	0.50	0.38	0.85

TABLE XII

F-7 NMI ACROSS THE FULL V1–V15, V17–V20 SWEEP ON THE LABELLED-SCENE PANEL. BOLD = BEST PER SCENE ACROSS ALL 19 RECIPES.

Scene	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14	V15	V17	V18	V19	V20
Indian Pines	0.34	0.35	0.43	0.25	0.20	0.25	0.16	0.43	0.10	0.08	0.25	0.42	0.29	0.31	0.26	0.18	0.27	0.21	0.44
Salinas	0.47	0.54	0.43	0.39	0.44	0.32	0.28	0.56	0.14	0.20	0.33	0.64	0.35	0.51	0.28	0.17	0.50	0.35	0.47
Salinas-A	0.62	0.64	0.68	0.37	0.56	0.33	0.16	0.52	0.20	0.28	0.42	0.55	0.49	0.55	0.51	0.34	0.52	0.33	0.68
Pavia U	0.47	0.38	0.55	0.38	0.14	0.43	0.23	0.54	0.02	0.00	0.21	0.61	0.22	0.49	0.28	0.21	0.52	0.36	0.54
Kennedy SC	0.41	0.40	0.54	0.32	0.25	0.34	0.20	0.17	0.11	0.17	0.22	0.51	0.31	0.43	0.29	0.21	0.28	0.25	0.52
Botswana	0.42	0.40	0.52	0.35	0.20	0.24	0.17	0.56	0.13	0.17	0.32	0.46	0.21	0.44	0.25	0.19	0.47	0.21	0.47
mean	0.45	0.45	0.52	0.35	0.30	0.32	0.20	0.46	0.12	0.15	0.29	0.53	0.31	0.46	0.31	0.22	0.43	0.29	0.52

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